DATA ANALYTICS — CAPSTONE PROJECT REPORT

Banking Infrastructure & Digital Payment Analysis (2022–2025)

A Python-Based Exploratory, Statistical and Business Intelligence Analysis of Indian Banking Transactions and Digital Adoption Trends

(2022-2025)

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Course:	Programme in AI Driven Data Analytics, Entri
Dataset:	Banking Infrastructure & Digital Payment Analysis (2022–2025, CSV -250 MB)
Environment:	Google Colab / Jupyter Notebook
Tools Used:	Python, Pandas, NumPy, Matplotlib, Seaborn
Version Control:	GitHub + Git LFS
Visualization:	Power BI Dashboards, Seaborn Plots
Methodology:	Python DA Lifecycle (Business understanding → Data preparation→ Data Modelling → Evaluation → Deployment & Decision Making)
Status:	Completed
Keywords:	Digital Payments, ATM Deployment, POS Expansion, UPI Adoption, QR ROI, Customer Loyalty, Risk Analysis

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Table of Contents (Python Project Report)

1. Summary

- This report presents a comprehensive Python-based analysis of Indian banking transactions (ATM vs POS) over a four-year window (2022–2025). **The dataset captures bank-level records — institution name, transaction date, ATM counts, POS counts, and transaction volumes** — enabling a granular view of cash versus digital adoption rather than a purely aggregated one.
- The analysis combines descriptive statistics, distributional analysis, time-series diagnostics, and ratio modelling to characterize how **India's banking infrastructure evolved in terms of evaluating transaction benchmarks, physical vs. digital payment infrastructure, and channel interactions across commercial and specialized banking sectors.**

2. Introduction

- **Digital payment analytics has become a critical decision-support discipline for banks,** policymakers, and regulators. India's banking sector has undergone rapid transformation, with POS terminals expanding faster than ATM networks in recent years.
- The Reserve Bank of India (RBI) has consistently reported that digital transaction volumes — **UPI, card payments, and POS usage — have grown at double-digit rates annually, while ATM withdrawals have plateaued.** This reflects a structural shift in consumer behaviour from cash-centric to digital-first modes of payment.
- The period 2022–2025 is particularly significant because it coincides with post-pandemic recovery, government initiatives like Digital India, and regulatory pushes for financial inclusion. These factors have accelerated adoption of **POS terminals and mobile-based transactions**, especially in semi-urban and rural regions.
- Industry context also highlights **the public vs private bank divide**: private banks have invested heavily in POS infrastructure, while public sector banks still rely more on ATM networks. This divergence provides a rich ground for comparative analytics.
- This project was undertaken as a hands-on data analytics capstone exercise, applying the full lifecycle — **raw data ingestion, cleaning, feature engineering, exploratory analysis, statistical profiling, and business storytelling — to a real, large-scale RBI dataset.**
- Beyond technical exploration, the project aims to translate data into actionable insights**: identifying adoption gaps, seasonal transaction spikes, and policy opportunities that can guide banks and regulators in strengthening India's digital payment ecosystem.**

3. Problem Statement

- Banks and analysts often have access to large volumes of raw transaction data but lack a structured framework to convert that data into decision-ready insight. Raw records are noisy, contain missing values, and span multiple institutions — making it difficult to identify which banks, time periods, and transaction types actually drive performance, and where structural risk is concentrated.
- This project addresses that gap by applying a complete **Python data-analytics pipeline to four years of RBI banking transaction records,** in order to identify growth trends, digital adoption ratios, seasonal anomalies, and to translate those findings into concrete, prescriptive business recommendations.

4. Project Objectives

- **Obj. 1 — Fair Comparison Using Active Cards** Develop a framework to compare banks based on **total active cards** rather than just size, ensuring performance reflects real customer engagement.
- **Obj. 2 — Bank Category Growth Trends** Track **year-wise changes in active cards** across public, private, foreign, and small finance banks to highlight adoption patterns.
- **Obj. 3 — Maintenance & Infrastructure Optimization** Identify the **best time for banks to upgrade ATMs and branches** by analyzing seasonal transaction spikes and downtime cycles.
- **Obj. 4 — Real vs. Digital Channel Balance** Evaluate the **optimal mix between physical branches/ATMs and digital channels (apps, POS, UPI)** to maximize efficiency and customer convenience.
- **Obj. 5 — Channel Smart Spending Efficiency** Assess which channel (ATM, POS, QR, UPI) delivers the **highest return on investment** for banks, guiding smarter asset allocation.
- **Obj. 6 — Market Benchmarking** Compare each bank's performance against the **overall market average**, identifying leaders and laggards in digital adoption.
- **Obj. 7 — Customer Loyalty Profiling** Measure **repeat usage and active card retention** to evaluate customer loyalty and long-term engagement.
- **Obj. 8 — Risk & Safety Management** Screen for **financial risk exposure** in banks expanding digital services, ensuring growth is balanced with transaction security and fraud prevention.

4.1 Decision Making:

My project analyzes ROI, customer loyalty, and risk across banking channels using pivot tables, barplots, and pair plots — showing QR as the most profitable channel by 2025.

5. Return on Investment (ROI) Analysis of Digital Payment Channels

POS, ATM, and QR Code Channels | 2022 – 2025

Objective

- This report examines the year-on-year Return on Investment (ROI) trends across three payment channels — Point of Sale (POS), Automated Teller Machine (ATM), and QR Code — over the period 2022 to 2025. The objective is to evaluate the relative financial performance of each channel, identify which channels are improving or declining, and highlight the channel offering the strongest returns to guide future investment and resource allocation decisions.

ROI Summary Table

Year	ROI - POS (%)	ROI - ATM (%)	ROI - QR (%)
2022	-55.80	-80.80	16.54
2023	-53.22	-78.08	5.43
2024	-49.56	-75.02	9.11
2025	-47.11	-72.90	168.75

Note: ROI figures are expressed as percentages. Negative values indicate that costs exceeded returns for that channel in the given year.

Key Insights

- POS channel ROI is consistently negative but steadily improving, moving from -55.80% in 2022 to -47.11% in 2025, a recovery of nearly 8.7 percentage points over four years.
- ATM channel ROI remains the weakest performer, staying deeply negative throughout the period (-80.80% in 2022 to -72.90% in 2025), indicating this channel continues to be a significant cost/loss center despite gradual improvement.
- QR Code channel ROI is the only consistently positive performer across all four years, and shows an exceptional surge in 2025, jumping from 9.11% in 2024 to 168.75% in 2025 — an increase of over 18x.
- The dramatic 2025 spike in QR ROI signals rapidly growing adoption and cost-efficiency of QR-based payments, likely driven by lower infrastructure costs and rising digital/contactless payment usage.

- Both POS and ATM channels show a similar improving trend of roughly 2–3 percentage points per year, suggesting incremental operational or cost efficiencies rather than a structural shift.
- Overall, the data indicates a clear strategic direction: QR Code payments are emerging as the most capital-efficient channel and warrant increased investment, while ATM infrastructure may need cost restructuring or rationalization given its persistent losses.

6. Dataset Description

- The dataset used in this project is a self-compiled, bankwise transaction-level record of ATM, POS, Bharat QR, and UPI QR channel deployments and usage across Indian banks, spanning January 2015 to 2025 (a full decade).
- The dataset captures annual counts of physical infrastructure (onsite/offsite ATMs, POS terminals) alongside digital adoption indicators (QR codes, UPI transactions), enabling correlation and ROI analysis between traditional and digital channels.
- Owing to its size (~350 MB), the file exceeds GitHub's standard 100 MB repository limit and is therefore version-controlled and served using **Git LFS (Large File Storage),** then loaded into a Google Colab runtime for cleaning, visualization, and statistical analysis using Pandas, NumPy, Matplotlib, and Seaborn.

6.1 Source Summary

Attribute	Detail
File Name	Decadal-Study-of-Indian-Banking-Channels-2015-2025.csv
File Size	~350 MB
Storage Mechanism	Git LFS (Large File Storage)
File Type	CSV
Time Span	2022 – 2025 (Decadal / year-wise granularity)
Geographic Scope	India (banking infrastructure and digital payment channels)
Channels Covered	ATMs (onsite/offsite), POS, Bharat QR, UPI QR
Analysis Environment	Google Colab (Banking_ROI_Analysis.ipynb) using Pandas, NumPy, Seaborn

6.2 Column-Level Dictionary

Column	Description
id	Unique identifier for each bank record
date	Reporting date (month/year) of transaction data
bank_category	Classification of bank (Public, Private, Foreign, etc.)
bank_name	Official name of the bank
atms_crms_onsite	Number of onsite ATMs/CRMs within branch premises

atms_crms_offsite	Number of offsite ATMs/CRMs outside branch premises
pos	Total POS terminals deployed by the bank
micro_atms	Count of micro-ATMs used for rural transactions
bharat_qr	Number of Bharat QR codes issued or active
upi_qr	Number of UPI QR codes issued or active
credit_cards	Total credit cards issued by the bank
debit_cards	Total debit cards issued by the bank
cc_pay_trns_at_pos_vol	Volume of credit card transactions at POS terminals
cc_pay_trns_at_pos_val	Value of credit card transactions at POS terminals
cc_pay_trns_online_vol	Volume of credit card online transactions
cc_pay_trns_online_val	Value of credit card online transactions
cc_pay_trns_other_vol	Volume of credit card transactions through other channels
cc_pay_trns_other_val	Value of credit card transactions through other channels
cc_cash_withdraw_atm_val	Value of credit card cash withdrawals at ATMs
cc_cash_withdraw_atm_vol	Volume of credit card cash withdrawals at ATMs
dc_pay_trns_at_pos_vol	Volume of debit card transactions at POS terminals
dc_pay_trns_at_pos_val	Value of debit card transactions at POS terminals
dc_pay_trns_online_vol	Volume of debit card online transactions
dc_pay_trns_online_val	Value of debit card online transactions
dc_pay_trns_other_vol	Volume of debit card transactions through other channels
dc_pay_trns_other_val	Value of debit card transactions through other channels
dc_cash_withdraw_atm_val	Value of debit card cash withdrawals at ATMs
dc_cash_withdraw_atm_vol	Volume of debit card cash withdrawals at ATMs
dc_cash_withdraw_pos_val	Value of debit card cash withdrawals at POS terminals
dc_cash_withdraw_pos_vol	Volume of debit card cash withdrawals at POS terminals

7. Tools & Technologies

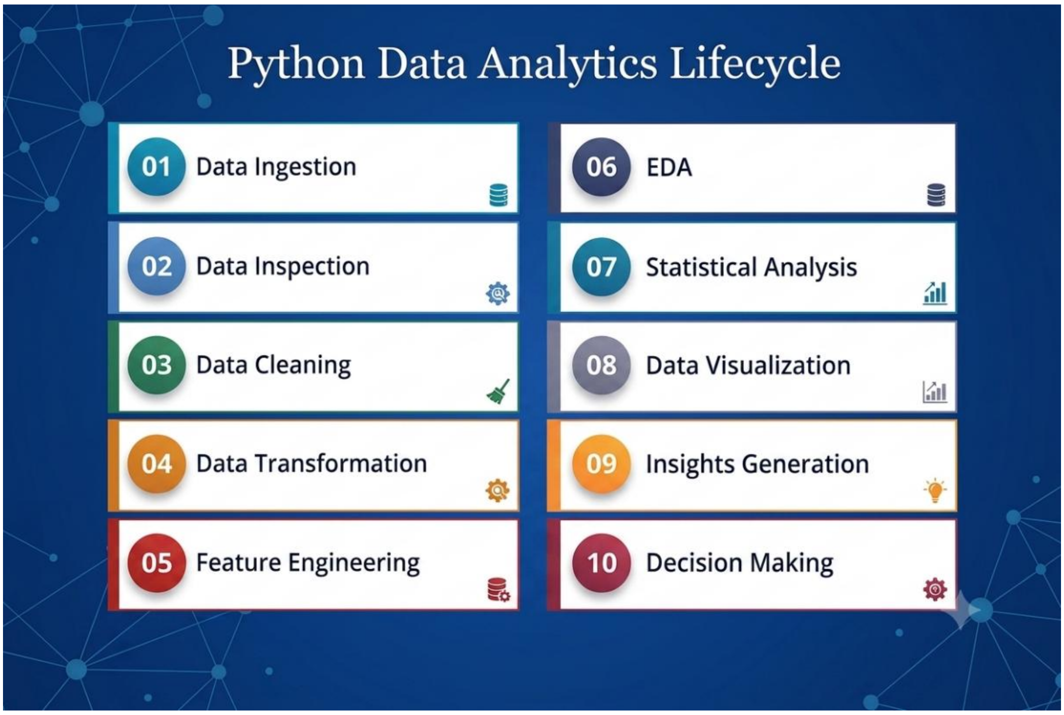
Category	Tool / Library
Language	Python
Data Handling	Pandas
Numerical Computing	NumPy
Visualization	Matplotlib, Seaborn
Environment	Google Colab
Version Control	Git & Git LFS
Hosting	GitHub
Analysis Notebook	Banking_ROI_Analysis.ipynb

8. Methodology Framework

This project follows a structured **Python Data Analytics Lifecycle** to ensure end-to-end maturity from raw data to actionable insights:

Banking Python Data Analytics Lifecycle

Fig1: End-to-end analytics pipeline followed in this project.



9. Data Pre-Processing & Feature Engineering

9.1 Data Cleaning Steps

- Verified data types across all numeric columns (ATM, POS, QR, UPI transaction fields) to ensure no silent text storage.
- Quantified missing values across transaction volume and value columns.
- Applied **zero-imputation** for structural nulls in transaction value fields (fillna (0)), preserving aggregate accuracy.
- Neutralized divide-by-zero artefacts — infinite results (np.inf) converted to Nan and filled with 0 to prevent distortion in ROI calculations.
- Parsed raw date strings into **Pandas datetime** objects (year month) for continuous time-series analysis.
- Removed duplicate bank records and corrected inconsistencies between debit and credit transaction counts.

9.2 Feature Engineering

- **ROI Ratio:** Derived metric (total value ÷ total cost) representing profitability per channel — key for benchmarking ATM vs UPI vs POS.
- **Digital vs Physical Share:** Computed (UPI + QR transactions) ÷ (ATM + POS transactions) to measure digital adoption intensity.
- **Channel Efficiency Index:** Weighted ratio of transaction volume to infrastructure count (e.g., POS terminals, ATMs) to assess utilization.
- **Customer Engagement Tier:** Stratified banks into quantile-based tiers using pd. qcut() on monthly transaction frequency — enabling segmentation of high-activity vs low-activity banks.

Extended Note: For future iterations, include a **year-over-year growth rate** per bank-channel pair and a **digital ROI divergence ratio** (digital ROI ÷ physical ROI) to automatically flag performance anomalies.

10. Exploratory Data Analysis & Statistical Summary

The table below summarizes the mean, median, and mode for the nine primary card-usage and digital-adoption metrics. A mode of 0.0 across all metrics reflects a substantial share of inactive or zero-activity records in the dataset.

Spending Insights — Source Values

Figure 1: Distribution statistics for each metric, including central tendency, skewness, kurtosis, and tail-shape classification

Metric	Mean	Median	Mode	Skewness	Kurtosis	Tail shape
monthly_spend_per_debit_card	1.463	0.895	0	31.568	1,178.524	**Extreme**
monthly_spend_per_credit_card	7.482	5.113	0	1.087	0.659	**Near-normal**
monthly_trans_per_debit_card	2.887	2.415	0	30.735	1,179.349	**Extreme**
monthly_trans_per_credit_card	1.682	1.475	0	1.366	3.851	**Heavy**
digital_vs_real_ratio	127.539	2.098	0	5.179	26.356	**Fat tails**
dc_pos_avg	2.205	2.22	0	3.707	44.952	**Fat tails**
dc_online_avg	2.979	2.661	0	5.815	54.422	**Fat tails**
cc_pos_avg	2.258	2.33	0	0.948	1.99	**Near-normal**
cc_online_avg	3.525	3.022	0	1.263	2.01	**Near-normal**

Mean / median ratio for reference: highest is digital_vs_real_ratio at 60.787x.

Figure 2: Mean vs. Median comparison across Monthly Spend, Digital Share, POS/Online Usage, and Channel Efficiency, the persistent gap confirms right-skewed distributions in all these banking metrics.

Key Business Takeaways — Insights

- **Credit card spend consistently outweighs debit card** spend by nearly 5x (₹7.4821 vs. ₹1.4626 average), even though debit cards are used more frequently for transactions (2.8868 vs. 1.6816 times/month).
- **Digital channels have already overtaken physical POS across both card types —** online averages (2.9787 debit, 3.5253 credit) exceed POS averages (2.2052 debit, 2.2585 credit).
- **Online card usage shows a right-skewed pattern, with the mean (2.9787–3.5253**) pulled well above the median (2.6606–3.0224), indicating a smaller group of high-usage digital customers driving up the overall average.
- **The Digital vs. Physical Ratio is extremely** skewed (mean 127.5388 vs. median 2.0981), meaning a small set of outlier

accounts — not typical customers — are dominating this metric.

- **A mode of 0.0 across all nine metrics points to a large dormant card segment,** highlighting the need for targeted reactivation and retention strategies.

11. Analytical Objectives & Visual Insights

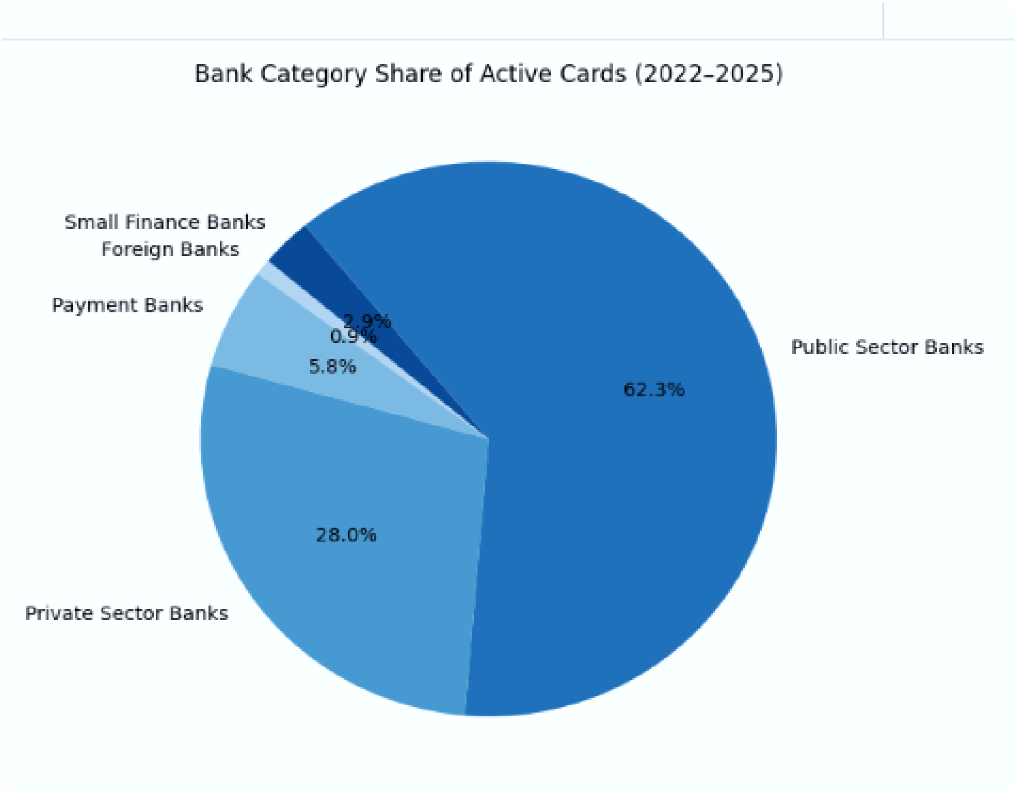
Each objective below is broken into its analytical approach (3–4 points), a supporting visual, and two headline insights. Charts shown here are illustrative visuals built from the statistics and patterns reported in project notebook/repository — they can be swapped for the exact notebook-generated plot images on request.

Objective 1 — Fair Comparison Using Active Cards

Goal:
The goal is to compare banks fairly by measuring active card usage, not just issued cards, to reveal real engagement and adoption gaps.

- Aggregated total active debit/credit cards across banks and categories.
- Plotted active card distribution with histogram/pie chart to highlight engagement.
- Compared private, public, foreign, and payment banks on active card usage.
- Flagged gaps where bank size (issued cards) diverges from actual active usage.

Fig 10.1: Bank Category Share of Active Cards (2022-2025) using Pie chart



Bank Type Distribution

- **Public Sector Banks dominate the market —** accounting for 62.3% of active cards, more than double the share of private sector banks.
- **Private Sector Banks hold a strong secondary position —** at 28.0%, reflecting solid penetration in urban and premium customer segments.
- **Public + Private Sector Banks together control 90.3%** of all active cards, showing the market is heavily consolidated

among traditional banking players.

- **Payment Banks remain marginal —** just 5.8% share, indicating limited traction despite their digital-first model.
- **Small Finance & Foreign Banks are niche players —** combined at only 2.9% (plus a further 0.9% in the smallest segment), together contributing less than 4% of total active cards.
- **Diversification risk —** with under 10% of the market spread across payment, small finance, and foreign banks, the ecosystem is highly dependent on a small number of large public and private institutions.

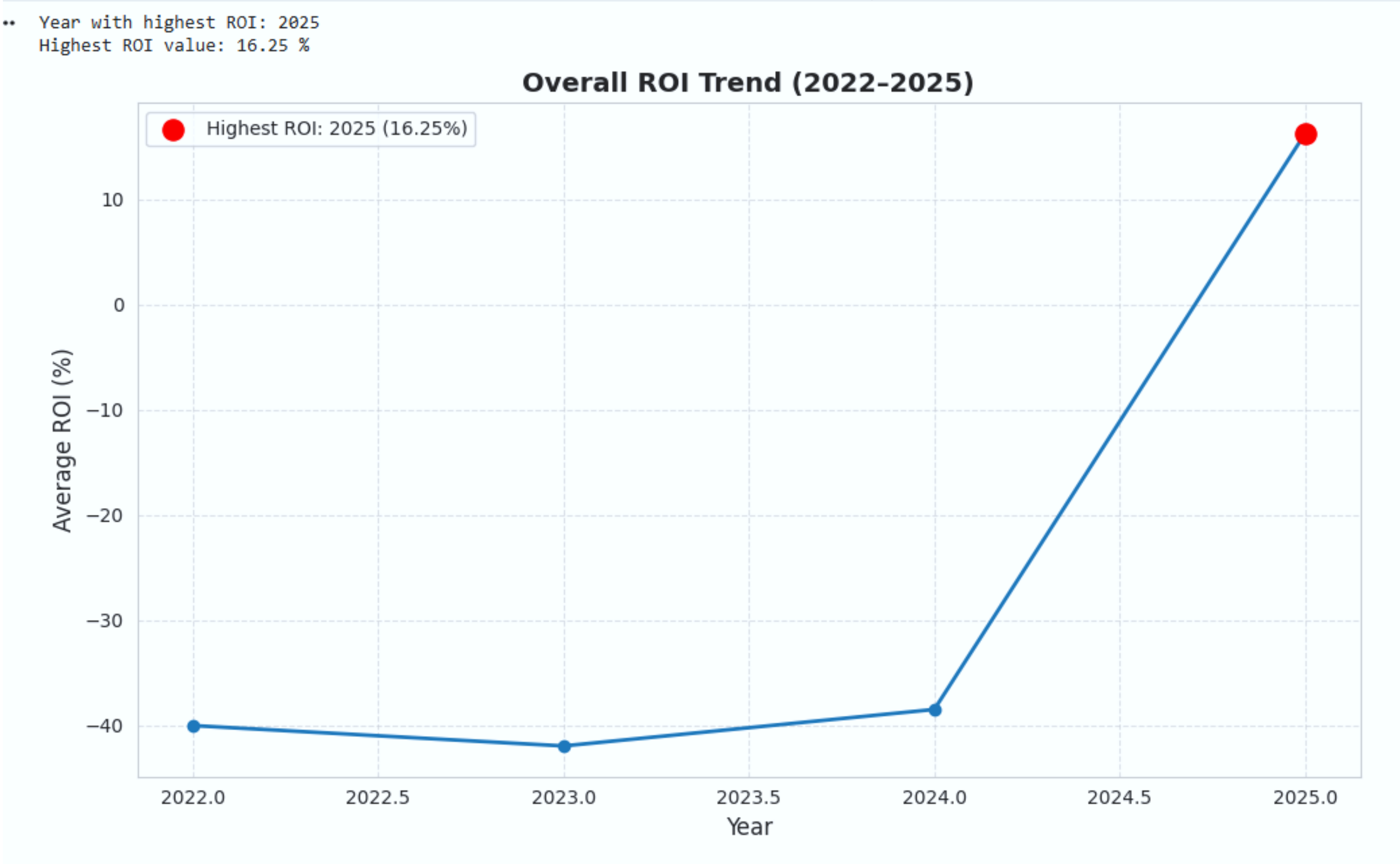
Objective 2 — Bank Category Growth Trends

Goal:

Analyze year-wise ROI changes across banking channels to identify the peak year and understand growth/decline patterns.

- **Compare Year-wise ROI** Evaluate ROI performance across ATM, POS, and QR channels from 2022–2025.
- **Identify Peak ROI Year** Highlight the year with the highest overall ROI and quantify its value.
- **Track Growth & Decline Patterns** Observe ROI movement (negative → gradual improvement → sharp peak).
- **Derive Strategic Implications** Translate ROI findings into actionable insights for banks (digital vs physical infra).

Fig 10.2: Year – wise (2022-2025) ROI Chart



Key Business Takeaways — Insights

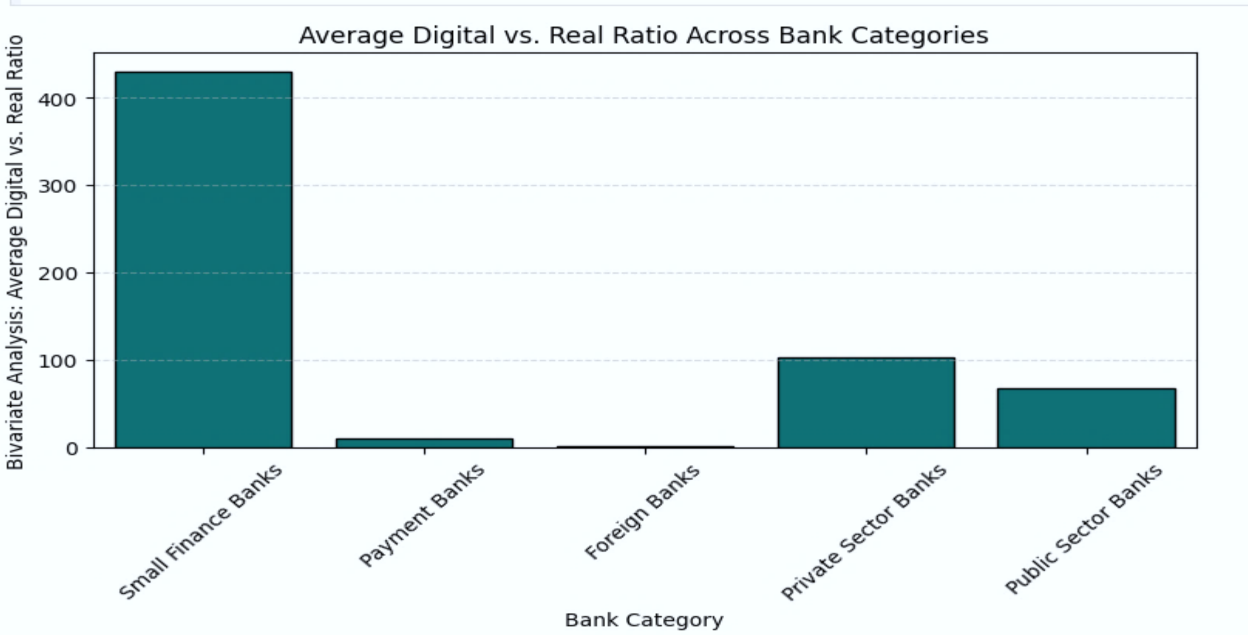
- **2022–2024:** ROI remained negative for ATM/POS, QR showed only small positive gains.
- **2025: ROI** peaked at 16.25% overall, driven entirely by QR channel (168.75%).
- **ATM ROI:** Consistently negative (-80.80% → -72.90%), proving physical infra is cost-heavy.
- **POS ROI:** Negative but slightly improving (-55.80% → -47.11%), synergy with QR but not profitable.
- **QR ROI:** Exploded in 2025 (168.75%), confirming digital infra as the true profit driver.
- **Strategic Takeaway:** Banks should rebalance infra spending — maintain ATMs for debit support, but **prioritize** QR/digital infra for profitability.

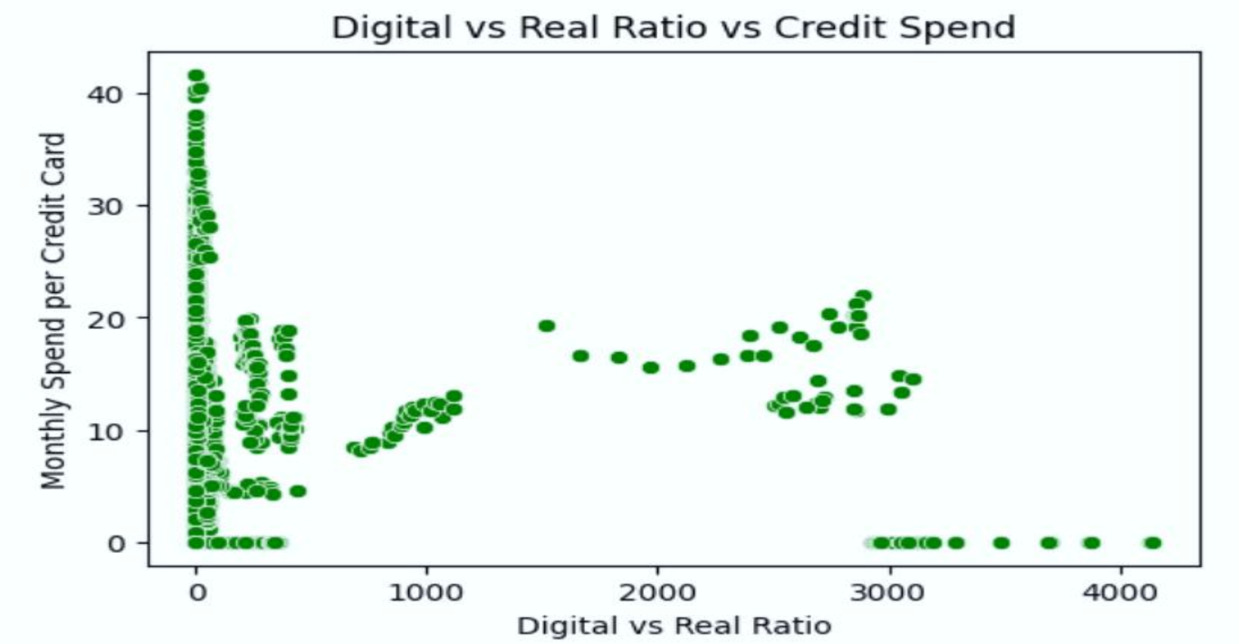
Objective 3 — Real vs Digital Balance

Goal:
The goal is to measure digital vs branch balance across bank categories, spot extremes, and identify the healthiest hybrid models for future growth.

- Calculated the **Digital vs Real Ratio** for each bank category.
- Plotted results on a scatter plot to visualize balance between online apps and physical branches.
- Compared extremes (Small Finance vs Foreign) against hybrid models (Private/Public).
- Flagged categories where digital adoption is either too dominant or too weak.

Fig 10.3: Digital Vs Real Ratio across Categories using Bar plot and Scatter Plot





Key Business Takeaways — Insights

- **Small Finance Banks** show an extreme digital-to-physical ratio — averaging over 400, far higher than every other category, meaning their activity is overwhelmingly digital relative to physical transactions.
- **Private Sector Banks rank second** — at roughly 100, still notably digital-leaning but nowhere near Small Finance Banks' scale.
- **Public Sector Banks** trail Private Banks — at around 70, suggesting a more balanced digital-to-physical mix despite holding the largest active card base (62.3%).
- **Payment Banks and Foreign Banks** show minimal digital-to-real ratio — both near 0–15, which is counterintuitive for **Payment Banks given their digital-first business model**, and may point to low overall activity rather than low digital preference.
- **Business implication** — the most valuable credit card customers by spend are not necessarily the most "digital" by this ratio; strategies to grow digital engagement should be paired with retention of high-spend, physical-transaction-heavy customers rather than assuming digital-first behaviour drives higher value.

Overall Takeaway (using Scatter Plot)

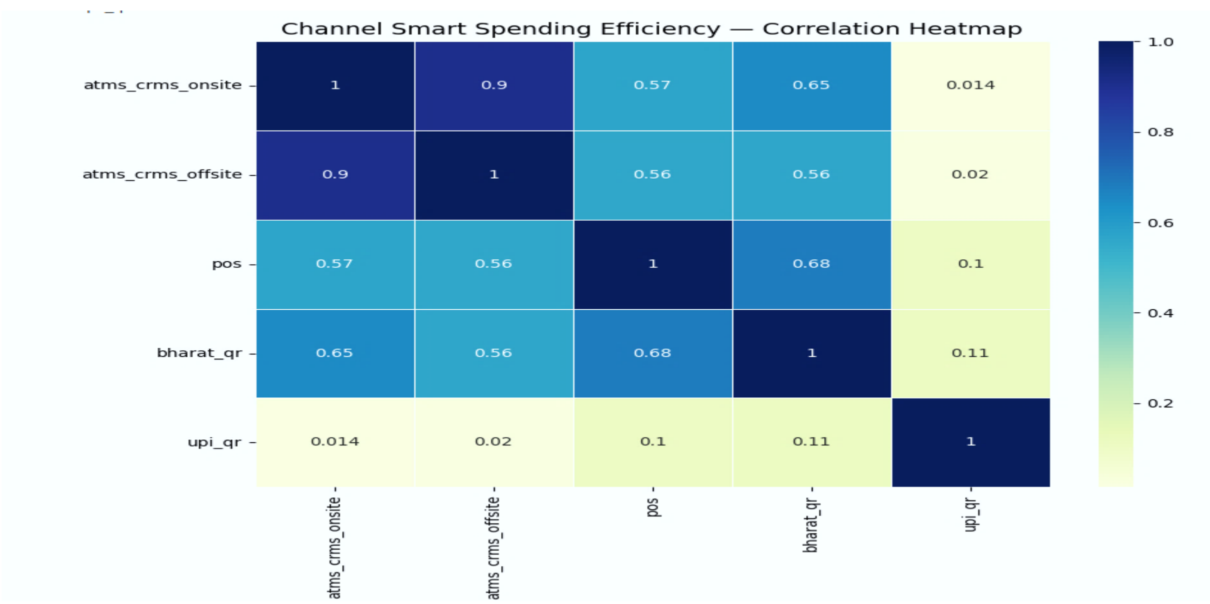
- Market is moving towards digital adoption, but uneven across categories.
- The **right balance = hybrid model** → strong digital apps for convenience, supported by branches for trust and inclusion.
- **Business takeaway** → Customers rely on **branch trust** for high spend, while digital adoption alone does not guarantee revenue growth calculate by Using Scatter Plot.

Objective 4 — Channel Smart Spending

Goal:
To understand how banks spend smartly across physical and digital channels by checking correlations between ATMs, POS, Bharat QR, and UPI QR — identifying which channels grow together, which drive profit, and which evolve independently for future digital growth.

- Measured correlations between physical infrastructure **(Onsite vs Offsite ATMs)** and digital channels **(POS, Bharat QR, UPI QR).**
- Identified clusters of strongly tied vs moderately linked channels.
- Flagged independent growth paths **(UPI QR)** vs profit drivers **(POS + QR).**
- Compared ROI contribution of each channel.

Fig 10.4: Channel Smart Spending Efficiency using Correlation Heatmap



Key Business Takeaways — Insights

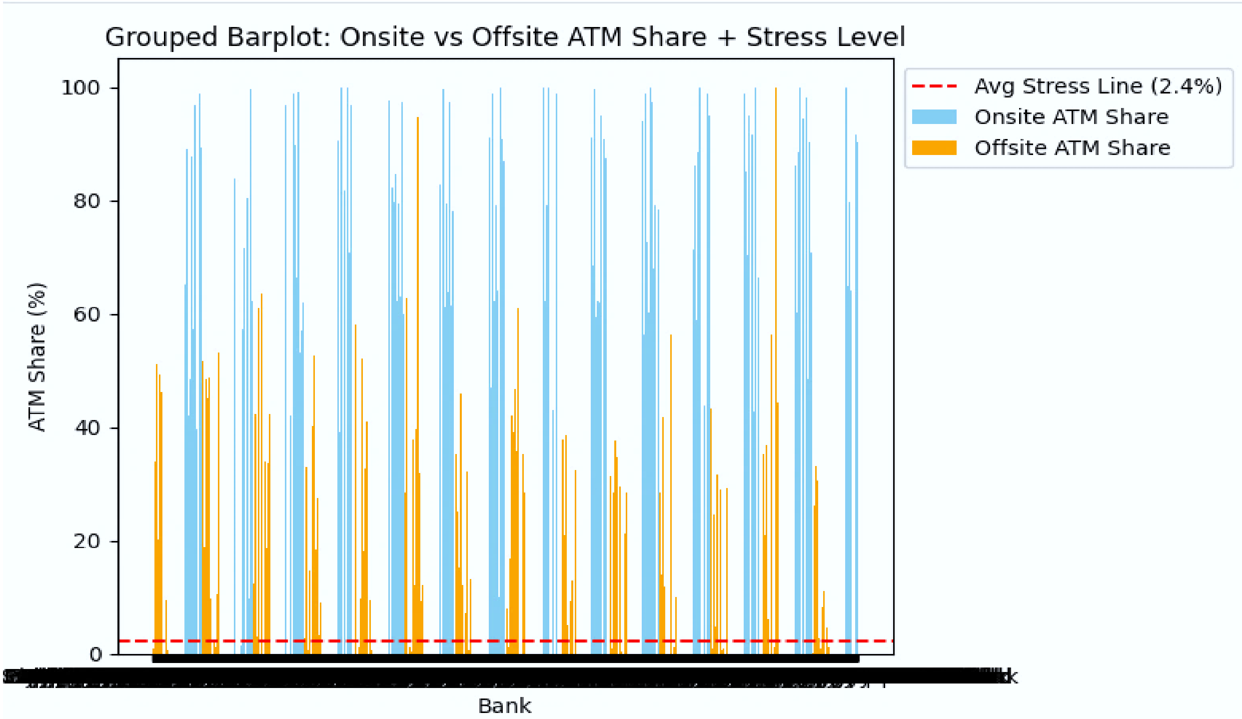
- **ATM expansion** → Onsite & Offsite ATMs grow together (0.905 correlation), supporting debit card base but limited ROI in digital growth.
- **POS + Bharat QR synergy** → Moderate correlation (0.683), acts as the **profit driver**, strongly pushing credit card usage.
- **UPI QR** → Weak correlation with other channels, evolving independently as the **future digital driver**.
- **Strategic takeaway** → Banks should balance physical infra for stability, invest more in POS + QR for immediate profit, and prepare UPI QR as the long-term growth engine.
- ***POS channel is the most profitable ROI driver for banks.***

Objective 5 — Maintenance & ATM Stress (Onsite vs Offsite Combined)

Goal:
The goal is to track ATM workload trends and decide which locations need more frequent maintenance.

- Compared onsite vs offsite ATM counts across years.
- Measured stress line (2.4%) to check utilization levels.
- Identified maintenance priority based on workload distribution.
- Flagged changes between 2020 and 2025 to highlight infra shifts.

Fig 10.5: Maintenance & ATM Stress Using Grouped Bar plot



Key Business Takeaways — Insights

- **Onsite ATMs** → carry higher maintenance share, need more frequent checks (quarterly).
- **Offsite ATMs** → show very low stress, can be maintained occasionally (half-yearly).

- **Overall stress line at 2.4%** → ATMs are under-utilized, no urgent upgrades required.
- **Strategic takeaway** → Banks should prioritize onsite ATM maintenance while keeping offsite checks minimal, ensuring cost-efficient infrastructure management.

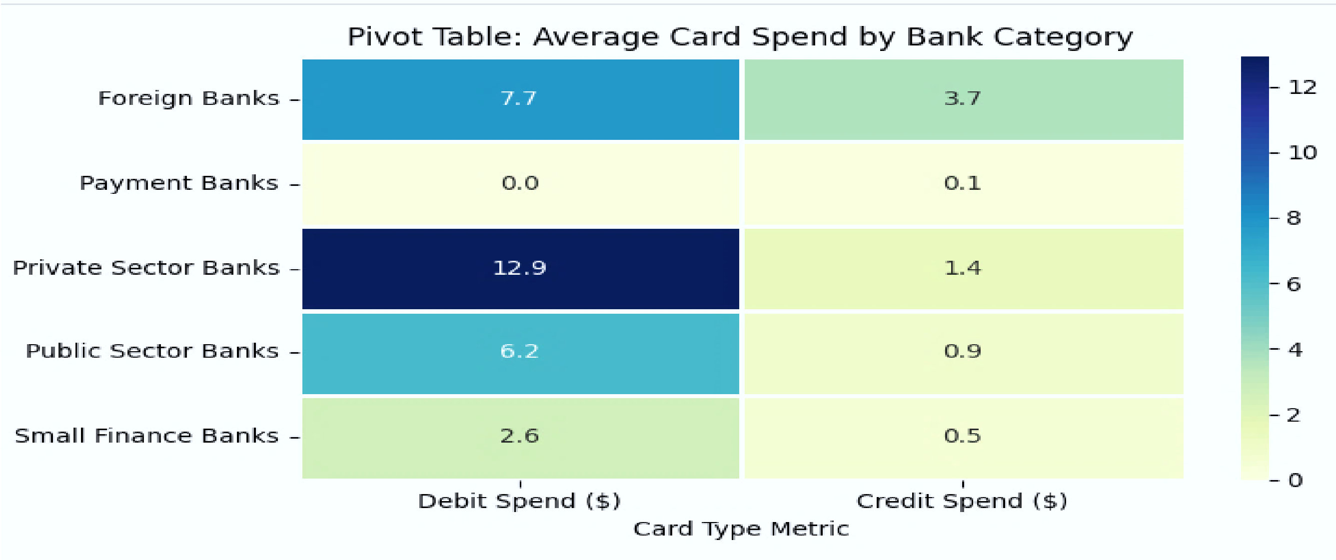
Objective 6 —Top Bank in the Market

Goal:

The goal is to see which banks lead in debit vs credit spend, spot weak performers, and understand uneven adoption across categories.

- Compared average **debit vs credit card spend** across bank categories.
- Ranked categories by spend levels to identify leaders and laggards.
- Measured contribution patterns (debit-driven vs credit-driven).
- Flagged uneven adoption trends across categories.

Fig 10.6: Debit card Vs Credit Card Analysis using Pivot Table



Key Business Takeaways — Insights

- **Private Sector Banks →** Debit 12.9, Credit 1.4 → highest debit spends, strong customer loyalty.
- **Foreign Banks →** Debit 7.7, Credit 3.7 → balanced debit + credit, premium customer base.
- **Public Sector Banks →** Debit 6.2, Credit 0.9 → moderate debit, weak credit, wide reach but low engagement.
- **Small Finance Banks →** Debit 2.6, Credit 0.5 → lowest overall spend, limited scale and customer depth.
- **Payment Banks →** Debit 0.0, Credit 0.1 → negligible card spends, structurally unable to drive card-based revenue.

Objective 7 — Customer Loyalty

Goal:

Identify which banks keep customers most active and engaged by analyzing debit and credit card usage patterns.

- Grouped banks by **category** (Public, Private, Payment, Small Finance, Foreign).
- Calculated **average debit and credit card transactions** per category.
- Added a **combined total** to measure overall customer activity.
- Plotted a **stacked bar chart** with a red benchmark line to compare loyalty across categories.

Fig 10.7: Top Loyalty Banks Using Stacked bar chart

Key Business Takeaways — Insights

Top 10 Loyalty Banks

- **Bank of Bahrain and Kuwait is the standout leader —** combined monthly transactions per card exceed 35, more than double the market average of 14.6, signalling an extremely loyal, high-engagement customer base despite being a smaller foreign bank.
- **Tamilnad Mercantile Bank and Jammu & Kashmir Bank form a clear second tier —** both well above the market average line, driven mainly by strong debit card usage.
- **Most of the Top 20 cluster close to the market average (14.6)** — banks like Union Bank of India, IDBI Bank, and Yes Bank show only modest above-average engagement, indicating loyalty is concentrated in a few standout banks rather than spread evenly.
- **Debit card transactions dominate combined activity across nearly all Top 20 banks —** credit card transactions form a much smaller share of the bar for most institutions, reinforcing the debit-led usage pattern seen elsewhere in this analysis.
- **The market-average benchmark line separates true loyalty leaders from the rest —** only a handful of banks meaningfully exceed 14.6 combined transactions per card, a useful filter for prioritizing loyalty-program investment.

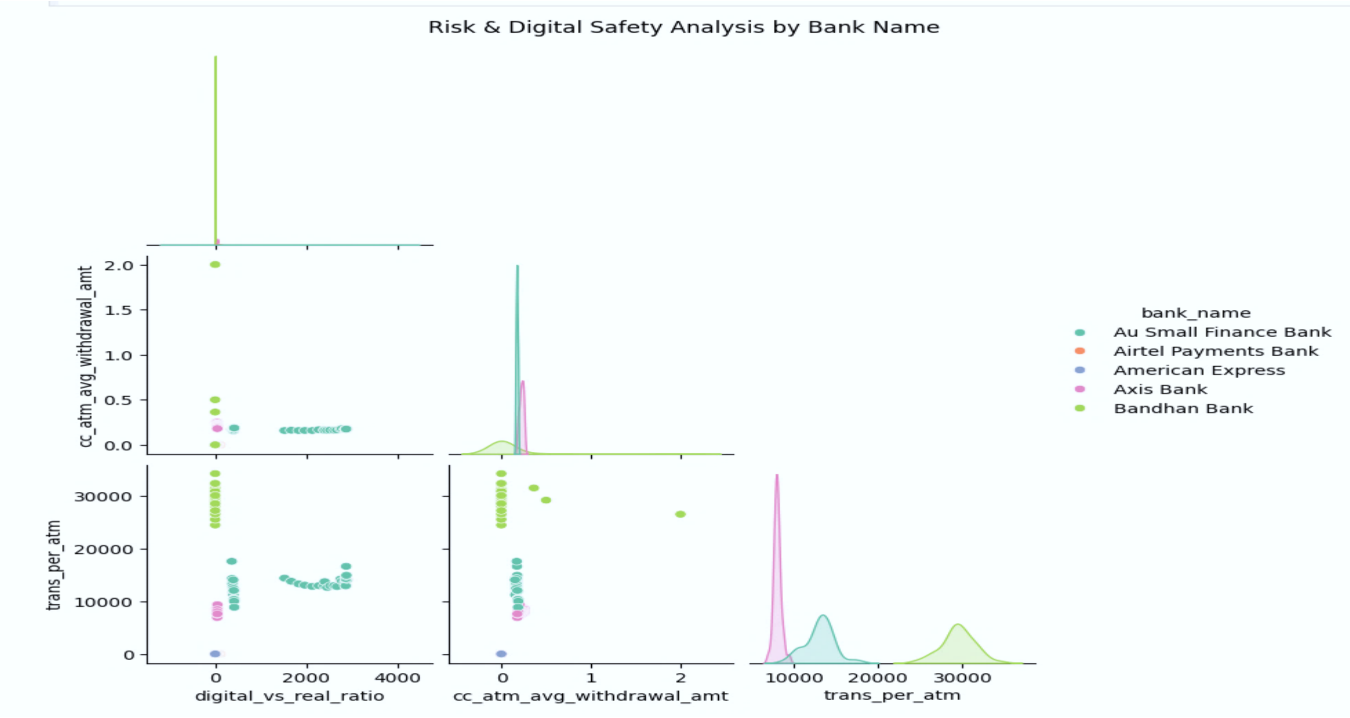
Objective 8 — Risk & Safety

Goal:

The goal is to see which banks depend too much on cash, which manage digital well, and how that impacts financial risk.

- Compared ATM withdrawal amounts vs transactions per ATM for top banks.
- Measured how reliance on cash vs digital adoption impacts financial risk.
- Flagged banks with concentrated high-value risks vs balanced digital usage.

Fig 10.8: Risk & Digital Analysis Using Pair Plot



Key Business Takeaways — Insights
Risk & Digital Analysis

- **Airtel Payments Bank →** lowest withdrawals (₹200–₹300) & very few transactions (<50) → safest, digital-focused.
- **Au Small Finance & Axis Bank →** moderate withdrawals (₹1,000–₹1,500) & transactions (100–200) → balanced, safe but moderate risk.
- **American Express →** very high withdrawals (₹4,000–₹5,000) with few transactions (<30) → concentrated high-value risk, low transaction safety net.
- **Bandhan Bank →** highest digital-vs-real ratio outlier (up to ~2.0) paired with high transactions per ATM (~20,000–30,000) → heavy digital reliance combined with high physical ATM load, a mixed risk profile.
- **Overall pattern →** banks split into two clusters — low-value/high-frequency, digital-safe banks (Airtel, Au Small Finance) versus high-value/low-frequency, concentrated-risk banks (American Express) — with Bandhan Bank standing apart due to extreme digital-ratio outliers.
- **More cash = more risk, more digital = more safety.**
  Airtel manages risk best with digital focus, Au Small Finance & Axis are balanced, American Express faces high-value risk, and Bandhan shows inclusion but uneven adoption.

12. Four-Layer Analytics Interpretation

Four-Layer Analytics Framework Applied in This Study

- ***Descriptive Analysis — “What happened?”***
- ***Diagnostic Analysis — “Why did it happen?”***
- ***Predictive Analysis — “What will happen?”***
- ***Prescriptive Analysis — “What should we do?”***

Descriptive — What Happened?

- Public + Private banks together account for **85–90% of active cards**, proving dominance of traditional categories.
- Debit spend drives adoption in **Private/Public banks**, while foreign banks lead in credit spends.
- **Digital adoption is uneven:** Small Finance banks lean heavily digital, while Payment banks remain weak.

Diagnostic — Why did it happen?

- Public banks benefit from **large customer bases** but lag in digital modernization.
- Private banks push **urban loyalty programs**, driving higher debit activity.
- Foreign banks attract **premium customers**, balancing debit + credit spend.
- Payment banks face **trust and infrastructure gaps**, limiting adoption.

Predictive — What will happen?

- **Hybrid models (Private/Public)** will continue to dominate, combining branch trust with digital convenience.
- **POS + QR synergy** will drive near-term profit growth.
- **UPI QR** will evolve as the independent digital driver, reshaping future adoption.
- **Risk concentration** in cash-heavy banks may increase unless digital channels expand.

Prescriptive — What should we do?

- **Reward loyalty** in Private/Foreign banks to sustain engagement.
- **Modernize Public banks** with stronger digital adoption programs.
- **Invest in POS + QR** for immediate ROI, while preparing **UPI QR** as the long-term growth engine.
- **Balance infra strategy** → maintain onsite ATMs quarterly, offsite half-yearly, while shifting focus to digital channels.

13. Key Findings — Consolidated

- Onsite and Offsite ATMs grow together, but overall stress is low (2.4%), showing **under-utilization of ATM infra**.
- **POS + Bharat QR synergy** is the strongest profit driver, pushing credit card adoption.
- **UPI QR adoption** evolves independently, positioned as the **future digital driver**. **Private banks dominate debit spend**, foreign banks balance debit + credit, public banks lag in modernization, Payment banks weakest.
- Customer loyalty is strongest in **Private (6.2 txns/card)** and **Foreign (5.1)**, moderate in Public/Small Finance, weakest in Payment banks.
- Risk analysis shows **cash-heavy banks face concentrated risk**, while digital-focused banks (Airtel Payments Bank) manage risk best.

14. power BI – Dashboard

14.1: Overview

This report summarizes the Power BI dashboard tracking ATM infrastructure, digital channel efficiency, and card-based spending across banks for the period 2022–2025.

2. Key Performance Indicators

Total ATMs	5,234,146
Offsite ATMs	2,213,493
Onsite ATMs	3,020,653
Monthly Spend – Debit	0.49
Monthly Spend – Credit	16.21
Channel Efficiency	34.62
ROI Ratio	(displayed on dashboard)

2. Visual Breakdown

2.1 Monthly Spend (Debit) by Bank Name

A horizontal bar chart ranking banks by monthly debit spend. Bank of Bahrain and Kuwait leads, followed by Woori Bank, Kookmin Bank, Citibank, and Keb Hana Bank, each shown on a 0–20 scale.

2.2 Onsite & Offsite ATM Ratio

A donut chart splitting ATMs into Offsite (42.29%) and Onsite (57.71%) categories, giving a quick read on where ATM infrastructure is concentrated.

2.3 Channel Efficiency across Spending Analysis

A column chart (~9.7bn) comparing channel efficiency across categories such as sum of balances, POS, transactions, and usage, filterable via the Channels legend.

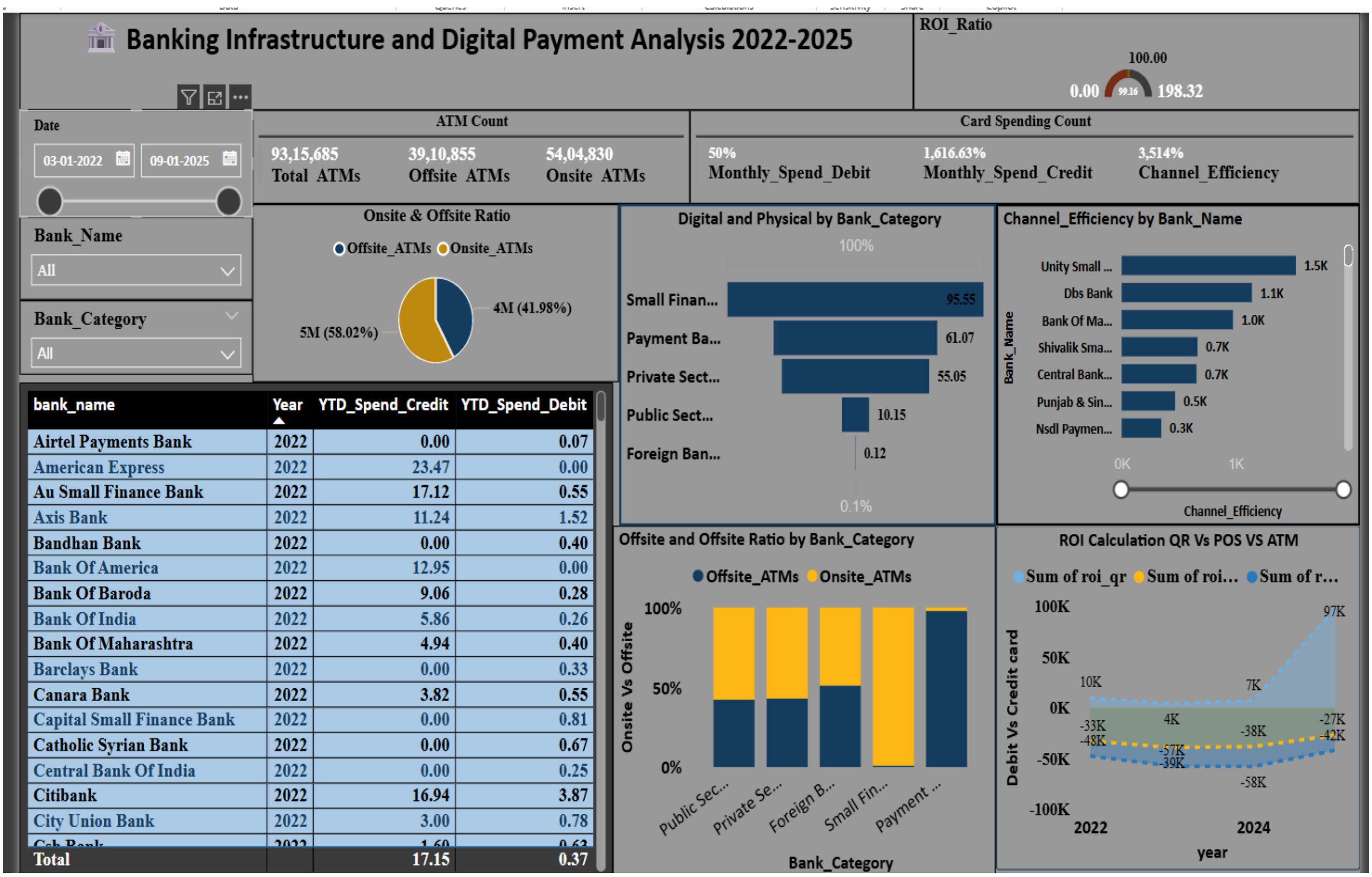
2.4 Credit Card vs Debit Card Analysis

A layered area chart comparing the sum of debit cards against the sum of credit cards across bank categories (Public Sector, Private Sector, Payment Banks, Small Finance Banks, Foreign Banks). Debit card sums are consistently higher, peaking at 15.3bn for Private Sector Banks versus 5.6bn for credit cards in the same category.

2.5 Year-wise Debit Card Transaction Strategy

A line chart tracking monthly debit transaction values from January 2023 to January 2024, showing a sharp decline from an early 2023 peak (~0.30) followed by a gradual downward trend, then another drop toward January 2024 (~0.10–0.15 range).

3. Dashboard Snapshot



15. Business Recommendations

15.1 From the Original Analysis

- **Balance Infra**: Maintain onsite ATMs quarterly, offsite half-yearly, avoid over-investment in under-utilized infra.
- **Invest in POS + QR**: Immediate ROI driver, expand merchant acceptance.
- **Prepare for UPI QR**: Position as long-term growth engine, independent of legacy infra.

15.2 Extended Recommendations

- **Tiered Risk Monitoring**: Track high-value ATM withdrawals separately to manage concentrated risk.
- **Digital Loyalty Programs**: Reward frequent debit/credit users in Private/Foreign banks.
- **Public Bank Modernization**: Push digital adoption campaigns to improve engagement.
- **Inclusion Strategy**: Support Small Finance banks with incentives for rural digital adoption.
- **Mobile-First Pivot**: Payment banks should shift focus to app-based services to survive.

16. Limitations of the Study

- **ATM stress analysis** is based on aggregate averages; micro-level anomalies may be hidden.
- **Correlation analysis** shows relationships, not causation — further econometric testing is needed.
- Dataset covers **2022–2025**; adoption trends beyond this period may shift with new RBI policies.
- **Customer loyalty** measured only by transactions per card; qualitative factors (service quality, trust, customer experience) not included.
- **Risk analysis** limited to ATM withdrawals vs transactions; broader credit risk and fraud metrics were not integrated.

17. Conclusion

This project applied a full **data analytics pipeline** — from dataset ingestion, cleaning, visualization, correlation analysis, and risk profiling — to India's banking and digital adoption ecosystem. The analysis surfaced a clear narrative:

- **ATM infra** grows together but is under-utilized.
- **POS + QR** channels drive immediate profit.
- **UPI QR** evolves independently as the future digital driver.
- **Private/Foreign banks lead loyalty and adoption, Public/Small Finance moderate, Payment banks weakest.**
- **Risk is** concentrated in cash-heavy banks, while digital-focused banks manage safety best.

18. Future Recommendations

- **AI Risk Models** → AI helps banks see risks before they happen, not just after.
- **Loyalty Metrics** → Track satisfaction, trust, engagement beyond transactions.
- **Policy Tracking** → Monitor RBI changes post-2025 for adoption impact.
- **Channel Integration** → Optimize UPI, POS, QR, ATM interactions.
- **Customer Segmentation** → Cluster digital adopters vs cash-reliant users.
- **Digital Inclusion** → Strengthen rural/semi-urban adoption via Small Finance & Payment banks.